ANSWER KEY — Integration Lesson 16 Test: Integrating Sine & Cosine

For the grader · full working shown · give credit for correct method with small arithmetic slips · missing + C on an indefinite integral costs half

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1. From the value table: what are $\sin(3\pi/2)$ and $\cos(2\pi)$?

> **Work:** $3\pi/2$ is three-quarters around the circle — the point is at the very BOTTOM: height −1, sideways position 0. So $\sin(3\pi/2) = -1$. 2π is a full lap, back to the start on the far right: $\cos(2\pi) = 1$. **Answer: $\sin(3\pi/2) = -1$ and $\cos(2\pi) = 1$.**

2. Find $\int \cos x \, dx$, and check by differentiating.

> **Work:** we need something whose derivative is $\cos x$, and $(\sin x)' = \cos x$ does it. Check: $(\sin x + C)' = \cos x$ ✓. **Answer: $\sin x + C$.**

3. Find $\int \sin x \, dx$. Why must the minus sign appear?

> **Work:** guess $\cos x$ fails, since $(\cos x)' = -\sin x$ (wrong sign). Flip the sign: $(-\cos x)' = -(-\sin x) = \sin x$ ✓. The minus must be there because differentiating cosine PRODUCES a minus sign, and we need a second minus already in place to cancel it. **Answer: $-\cos x + C$.**

4. Find $\int 5 \sin x \, dx$.

> **Work:** the constant 5 rides along: $5 \cdot (-\cos x) + C$. Check: $(-5 \cos x)' = -5 \cdot (-\sin x) = 5 \sin x$ ✓. **Answer: $-5 \cos x + C$.**

5. Find $\int (\cos x - \sin x) \, dx$, and check by differentiating.

> **Work:** term by term: $\int \cos x \, dx = \sin x$, and $\int (-\sin x) \, dx = -(-\cos x) = +\cos x$. Sum: $\sin x + \cos x + C$. Check: $(\sin x + \cos x)' = \cos x - \sin x$ ✓. **Answer: $\sin x + \cos x + C$.** (Common error: $\sin x - \cos x$, whose derivative is $\cos x + \sin x$ — wrong.)

6. Compute $\int_0^{\pi/2} \sin x \, dx$.

> **Work:** antiderivative $-\cos x$. FTC: $[-\cos x]_0^{\pi/2} = (-\cos(\pi/2)) - (-\cos 0) = (-0) - (-1) = 0 + 1 = 1$. **Answer: 1.**

7. Compute $\int_0^{\pi} \cos x \, dx$ and explain via signed area.

> **Work:** $[\sin x]_0^{\pi} = \sin \pi - \sin 0 = 0 - 0 = 0$. Explanation: $\cos x$ is ABOVE the axis on $[0, \pi/2]$ (contributing +1) and BELOW it on $[\pi/2, \pi]$ (contributing −1); the two pieces cancel exactly. **Answer: 0 — equal areas above and below the axis cancel.**

8. Compute $\int_{\pi}^{2\pi} \sin x \, dx$. What does the sign tell you?

> **Work:** $[-\cos x]_{\pi}^{2\pi} = (-\cos 2\pi) - (-\cos \pi) = (-1) - (-(-1)) = -1 - 1 = -2$. The negative sign says the sine curve lies BELOW the x-axis on this whole interval (it's the downward hump); the region has size 2 but counts as −2 in signed area. **Answer: −2; sine is below the axis there.**

9. Average value of cos x on $[0, \pi/2]$.

> **Work:** average value = (integral) ÷ (interval length). Integral: $\int_0^{\pi/2} \cos x \, dx = [\sin x]_0^{\pi/2} = 1 - 0 = 1$. Length: $\pi/2$. Average $= 1 \div (\pi/2) = 2/\pi \approx 0.64$. **Answer: $2/\pi$ (≈ 0.64).**

10. Challenge: compute $\int_0^{\pi} (2 \sin x + x) \, dx$.

> **Work:** antiderivative term by term: $\int 2 \sin x \, dx = -2 \cos x$ and $\int x \, dx = x^2/2$, so $F(x) = -2 \cos x + x^2/2$. (Check: $F' = 2 \sin x + x$ ✓.) Top limit: $F(\pi) = -2 \cos \pi + \pi^2/2 = -2(-1) + \pi^2/2 = 2 + \pi^2/2$. Bottom limit: $F(0) = -2 \cos 0 + 0 = -2$. Subtract: $(2 + \pi^2/2) - (-2) = 4 + \pi^2/2$. **Answer: $4 + \pi^2/2$ (≈ 8.93).**